

- 6 (a) Figure 6.1 shows a transmission electron micrograph of part of a chloroplast of Timothy grass, *Phleum pratense*.

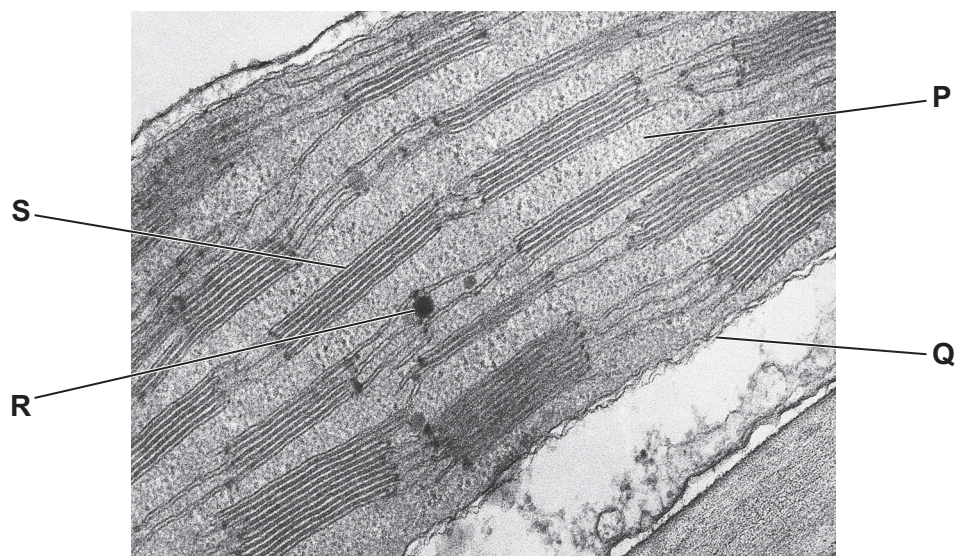


Figure 6.1

State the letter **P**, **Q**, **R** or **S** from Figure 6.1 that identifies:

- a structure containing photosynthetic pigments
- an area where the light-independent stage of photosynthesis is carried out

[2]

(b) A student carried out an experiment to investigate the effect of light intensity on the rate of photosynthesis in the brown alga *Fucus serratus*. Brown algae are aquatic multicellular protists.

- Discs were cut from the photosynthetic tissue of *F. serratus*.
- Ten discs were added to each of four beakers containing 50 cm³ of sea water.
- The beakers were illuminated with identical LED lamps placed at different distances (d), in centimetres, from the beakers.
- The relative light intensity was determined by calculating $1/d^2$.
- When the discs, which were denser than sea water, had sunk to the bottom of the beakers, the student started a timer.
- The student observed the discs rising to the surface of the water in the beakers and recorded the time taken (t), in minutes, for the fifth disc from each beaker to reach the surface.
- The rate of photosynthesis was determined by calculating $1000/t$.

The results are shown in Table 6.1.

Table 6.1

distance (d) of beaker from lamp / cm	relative light intensity	time (t) for fifth disc to reach the surface / min	rate of photosynthesis / min ⁻¹
5	0.0400		43.5
10	0.0100	36	27.8
15		52	19.2
20	0.0025	88	11.4

(i) Complete Table 6.1 to show:

- the time for the fifth disc to reach the surface of the water in the beaker when the distance (d) between the beaker and lamp was 5 cm
- the relative light intensity ($1/d^2$) when the distance between the beaker and lamp was 15 cm.

[2]



(ii) Use the data in Table 6.1 to:

- describe the relationship between light intensity and the rate of photosynthesis
- explain why the discs rose to the surface of the water in the beakers.

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(iii) The student found that there was no increase in the rate of photosynthesis when two identical lamps were placed 5 cm from the beaker.

Suggest why there was no increase in the rate of photosynthesis.

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..... [2]

[Total: 10]